

Chapter 46 — Neuromuscular Dysfunction of the Lower Urinary Tract in Children

IF YOU READ NOTHING ELSE

- **Detrusor leak point pressure (DLPP) >40 cm H₂O** → vesicoureteric reflux (VUR) / upper-tract damage (McGuire 1981)
- **Hostile ≥40; intermediate 25–39; safe <25 cm H₂O**
- **Clean intermittent catheterisation (CIC) + antimuscarinic early** = standard for hostile neonatal bladder
- **Expectant vs early:** end-stage renal disease (ESRD) ~30% vs 0–1.6%
- **Urodynamics (UDS) within 3 months;** after spinal cord injury (SCI) only **once spinal shock resolves (3–4 months)**
- **Folic acid:** 400 µg all women; 4–5 mg if affected; start ≥4 weeks pre-conception
- **Dilating VUR (III–V):** 7% of spina bifida (SB) neonates; **avoid Credé**
- **Botulinum toxin A (BTA):** 10 U/kg, max 360 U, ~30 sites, spare trigone
- **Dilating VUR at augmentation** → **reimplant simultaneously**
- **Autonomic dysreflexia: lesion ≥T6** → monitor blood pressure (BP), **α-blocker ready** during UDS

NEURAL TUBE DEFECTS

- **Commonest cause** of paediatric neuropathic lower urinary tract (LUT) dysfunction: **neural tube defect (NTD); myelomeningocele** is commonest and most severe
- **Cut-offs:** ~10% of SB deaths urological; survival ~90% at 10 years; first-year mortality ~5%
- **Strongest infant mortality predictors:** preterm (RR ~4.5), low birth weight (RR ~4.8)
- **Neural tube closure:** day 18 → 35, cephalad → caudal
- **Endoscopic third ventriculostomy + choroid plexus cauterisation:** >70% shunt-free
- **Myelomeningocele level:** lumbosacral 47%, lumbar 26%, sacral 20%, thoracic 7%
- **Bony level ≠ neurological level** (±1–3 vertebrae) → **UDS, not anatomy**
- **Sacral dimple, hair tuft, vascular malformation, abnormal cleft** in ~90% occult dysraphism
- **SB occulta** up to 15% population — **normal variant**
- **Recurrence:** 5% after one affected pregnancy; ≤10% after two
- **Arnold–Chiari II:** almost all open NTDs
- **Folic acid:** supplements ↓ 50–70%; fortification ↓ 20–50%
- **Folate antagonists:** valproate, carbamazepine
- **MOMS (Management of Myelomeningocele Study):**
 - prenatal repair <26 weeks: death/shunt RR (relative risk) 0.70; 40% still shunted
 - 34.1 vs 37.3 weeks; 13% <30 weeks; ½ uterine dehiscence/thinning
- **Newborn needle EMG:** ~40% intact / ~25% partial / ~35% complete denervation
- **UDS after SCI:** ≥6 weeks (International Children's Continence Society, ICCS)
- **UMPIRE (CDC urologic and renal protocol) baseline:**
 - ultrasound 56% normal / 40% mild / 4% moderate–severe
 - DMSA defects 8%
- **Ultrasound:** 6, 12, 18, 24 months, then yearly
- **Spinal ultrasound before posterior arch ossification** (first months)
- **Cystatin C > creatinine** in SB (muscle-independent); **creatinine baseline still advised**
- **Vitamin B12 yearly from 2 years after reconstruction**
- **Post-augmentation cystoscopy:** EAU (European Association of Urology) annual from 10–15 years
- **No antibiotic prophylaxis on CIC;** treat symptomatic UTI only
- **Vesicostomy = temporary bridge; augmentation = definitive; ileum preferred** (British Association of Paediatric Urologists, BAPU)
- **SBA UTI (<3 years):** urinalysis + catheter culture + **fever ≥38.0 °C**
- **Receptors:** M3 mediates contraction; M2 more numerous
- **First-line:** CIC (Lapides 1971) + antimuscarinic; oxybutynin commonest
- **Oxybutynin:**
 - oral 0.2 mg/kg/dose up to three times daily, max 5 mg/dose
 - patch 3.9 mg/day, ~20% stop (skin)
 - intravesical 0.1–0.9 mg/kg/day

NEUROPATHIC LOWER URINARY TRACT DYSFUNCTION

- **Management priority:** kidneys first → continence → urinary tract infection (UTI) → sexual function
- **Normal compliance** >15 mL/cm H₂O
- **UDS timing:**
 - safe from 3 months
 - **SBA (Spina Bifida Association):** annually to 36 months
 - 2–3 months after changing antimuscarinic
 - closed NTD/tumour: before + 3–4 months after surgery
- **Gold standards:**
 - **video UDS** for cystography
 - **needle electromyography (EMG)** for detrusor–sphincter dyssynergia (DSD)
 - **dimercaptosuccinic acid (DMSA)** for cortical defects
 - **clearance study** for glomerular filtration rate (GFR)
- **Patch EMG unreliable** → DSD dropped from UMPIRE high-risk category
- **FDA (US Food and Drug Administration) paediatric approval:**
 - oxybutynin ≥5 years; solifenacin ≥2 years; mirabegron ≥3 years
 - fesoterodine ≥6 years and >25 kg
 - vibegron: no
- **Trospium:** quaternary, half-life 8 vs 2 hours (oxybutynin), few CNS effects
- **Fesoterodine = tolterodine metabolite** (5-hydroxymethyl tolterodine); less variability
- **β3 agonist side effect:** hypertension
- **Sympathomimetics:** bladder-neck α-receptors; unlicensed; retention if voiding
- **Early vs expectant:**
 - scarring 4–12% vs 27–50%
 - hydronephrosis 0–15% vs 18–80%
 - UTI 0.3/year vs ~50% by 15 months
 - 18–24% fewer augmentations
- **No benefit** of early intervention if empties spontaneously

- **BTA:**
 - continence **65–87%**; lasts **6–10.5 months**
 - minimum age **2 years**
 - **~90% avoid augmentation**
 - dose reduction → relapse
- **VUR:**
 - **30–40% by 5 years** untreated
 - **resolves 30–55%** with CIC + antimuscarinic
- **Grade I–III and emptying → prophylaxis only; IV–V or not emptying → CIC**
- **Bacteriuria on CIC: harmless unless high-grade VUR**
- **Credé: harmful with reactive sphincter; harmless if denervated**
- **Operate (antireflux) if:**
 - febrile UTI despite prophylaxis + CIC
 - persistent hydronephrosis despite emptying
 - ureterovesical junction anomaly
- **Antireflux success ~95% since CIC (Jeffs 1976); unilateral VUR → unilateral surgery**
- **Endoscopic vs open: 61–73% vs 84–96%; 40% recur at 4.5 years; ~95% with BTA**
- **spinal ultrasound <3 months**
- **UDS only if bony spine anomaly, cord defect, or VCUG/ultrasound signs**
- **UDS before definitive surgery → after pull-through; ESRD ~2%**
- **Syndromic ARM: VACTERL, MURCS, OEIS, Currarino**
- **Currarino: ARM + sacrococcygeal defect + presacral mass; tethered >80%**
- **Peña posterior sagittal anorectoplasty (PSARP): nerve-sparing; not the cause of neuropathic bladder**
- **Krickenbeck: ARM classification by fistula site**
- **Sacrococcygeal teratoma: 35–45% neuropathic → UDS before + after surgery, all**
- **Rhabdomyosarcoma:**
 - **frequency–volume chart all survivors**
 - **radiotherapy → high risk**
 - **~20% reconstruction/diversion**
- **Hirschsprung: UDS only if symptomatic, 6 months post-op**
 - **unstable detrusor ≤45%, often resolves after resection**
- **Nerve injury:**
 - **parasympathetic (pelvic splanchnic) → underactive**
 - **sympathetic (hypogastric) → poor compliance**

SEXUAL FUNCTION

- **Central precocious puberty: ~50% girls, 10–30% boys**
- **Erectile dysfunction (ED) 75% (≥18 years); sildenafil ~80% effective (randomised controlled trial)**
- **Male reproductive function: S1 or lower → normal; above → ~50%**
- **Female sexual dysfunction 55.5%**
- **70–80% of women conceive, uneventful pregnancy; caesarean common**
- **Offspring SB 3.7%; both parents ~15%**
- **Absolute contraindications: intrauterine device; latex with allergy**
- **Relative: combined oestrogen–progestin → thromboembolism if immobile**

OTHER CONDITIONS ASSOCIATED WITH NEUROPATHIC LOWER URINARY TRACT DYSFUNCTION

- **Cerebral palsy (CP):**
 - **commonest childhood physical disability; ~3.3/1000**
 - continence late
 - median continence **3.6–4.1 years** (diplegia/hemiplegia, good intellect) **vs 10–13 years** (tetraplegia)
 - **upper tract <5%; UDS ½ normal; DSD ~12%**
- **SCI: adolescents 6× children; spinal shock → underactive bladder, non-reactive sphincter**
- **Suprasacral SCI → overactivity + DSD; cauda equina → no return**
- **Transverse myelitis (TM):**
 - **retention >95% acute; areflexia ~70% acute**
 - **spinal shock ≤6 weeks**
 - **57–73% need CIC**
 - **>½ on CIC still need it at 1 year**
 - **~1400/year (USA), 28% children; peaks 10–19 years**
 - acute treatment: **steroids, plasma exchange, intravenous immunoglobulin**
- **Anorectal malformation (ARM):**
 - **1 in 5000**
 - urinary anomalies **high 70% vs low 35%**
 - **cord anomaly 18–50%; neuropathic 18–35%**

CLASSIC TRAPS

- **Trap: bony level predicts bladder — Actually: unpredictable; do UDS**
- **Trap: normal creatinine = normal GFR in SB — Actually: low muscle mass masks loss; use cystatin C**
- **Trap: high-grade VUR with "safe" pressures — Actually: VUR pops off pressure; falsely reassuring**
- **Trap: patch EMG diagnoses DSD — Actually: unreliable (UMPIRE); needle EMG**
- **Trap: repeat BTA causes fibrosis or tachyphylaxis — Actually: neither; fibrosis falls**
- **Trap: augmentation alone cures dilating VUR — Actually: persists 20–47%; reimplant simultaneously**
- **Trap: antimuscarinics impair cognition — Actually: no effect (randomised double-blind trial)**
- **Trap: PSARP causes ARM neuropathic bladder — Actually: associated cord anomaly**
- **Trap: MOMS abolished shunting — Actually: 40% still shunted; more prematurity**
- **Trap: female fertility impaired in SB — Actually: hormonal control, preserved**